



Alert Triage with Threat Intelligence Report

1. Introduction

Alert triage is a critical process in Security Operations Centre (SOC) operations that involves analyzing security alerts to determine whether they represent genuine threats or false positives. Security monitoring systems generate large volumes of alerts, making it essential to validate and prioritize them using threat intelligence and contextual analysis.

In this task, a simulated alert related to suspicious PowerShell execution was analyzed using **Wazuh SIEM**. The alert was validated using external threat intelligence platforms including **VirusTotal** and **AlienVault OTX** to verify indicators of compromise (IOCs). The analysis helped determine the authenticity of the threat and supported incident response decisions.

This task demonstrates how threat intelligence enrichment improves detection accuracy and enables faster response to security incidents.

2. Objective

The objective of this task is to perform alert triage by analyzing security alerts and validating indicators of compromise using threat intelligence sources.

Key Objectives

- Analyze security alerts generated by SIEM systems.
 - Validate indicators of compromise using threat intelligence.
 - Identify false positives and real threats.
 - Assign appropriate severity levels to alerts.
 - Support incident response decision-making.
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3. Tools Used

- Wazuh Security - monitoring and alert detection
 - VirusTotal - IOC reputation analysis
 - AlienVault OTX - Threat intelligence validation
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4. Alert Triage Process Overview

Alert triage involves several steps to evaluate the authenticity and severity of security events.

Alert Triage Steps

1. Review alert details and context.
2. Identify indicators of compromise.
3. Validate indicators using threat intelligence.
4. Assess severity and impact.
5. Determine response actions.

This structured process helps reduce false positives and improves SOC efficiency.

5. Methodology / Implementation

5.1 Alert Identification

A simulated high-priority alert related to suspicious PowerShell execution was generated in the SIEM environment. PowerShell-based attacks are commonly used by attackers to execute malicious commands, perform lateral movement, and establish command-and-control communication.

Alert Characteristics

- Alert Type: Suspicious PowerShell Execution
- Detection Source: Endpoint monitoring
- Suspicious Behavior: Unauthorized script execution
- Associated IP Address: 192.168.1.101

5.2 Alert Analysis

The alert details were reviewed to determine potential risk and system impact.

Alert Triage Table

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.101	High	Open

The alert was assigned high priority due to its association with possible malicious activity.



5.3 Indicator of Compromise (IOC) Validation

The source IP address associated with the alert was validated using external threat intelligence platforms.

Validation Process

- Checked IP reputation using VirusTotal.
- Cross-referenced threat intelligence data using AlienVault OTX.
- Analyzed detection history and reputation scores.

IOC Validation Results

The IP address 192.168.1.101 was analyzed using VirusTotal and AlienVault OTX. Multiple security vendors flagged the IP as malicious and associated it with botnet and malware distribution activity. Threat intelligence reports linked the IP to command-and-control infrastructure, confirming the alert as a genuine security incident requiring investigation.

5.4 Threat Classification

The suspicious PowerShell activity was mapped to attacker behavior defined in the **MITRE ATT&CK framework**.

Threat Mapping

- Technique: Command execution via scripting.
- Attack Category: Execution and persistence.
- Risk Level: High.

Mapping the activity to standardized techniques helps improve detection and response.

5.5 Incident Response Decision

Based on alert validation and threat intelligence findings, the incident required escalation for further investigation.

Recommended Actions

- Investigate endpoint activity.
- Isolate affected system.
- Block malicious IP address.
- Monitor related network traffic.
- Perform forensic analysis.



6. Results and Findings

The alert triage process produced the following outcomes:

- Successful identification of suspicious activity.
- Validation of malicious IP address.
- Accurate classification of alert severity.
- Reduction of false-positive risk.
- Improved confidence in incident response decisions.

Threat intelligence enrichment significantly enhanced detection accuracy.

7. Security Impact

Effective alert triage improves organizational security posture by:

- Detecting threats at an early stage.
 - Reducing alert fatigue.
 - Prioritizing high-risk incidents.
 - Improving response efficiency.
 - Strengthening proactive defense capabilities.
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8. Conclusion

This task demonstrated the process of alert triage using SIEM monitoring and threat intelligence validation. The simulation showed how security alerts are analyzed, validated, and prioritized based on contextual information and external intelligence sources. Effective alert triage enhances detection accuracy, reduces false positives, and supports efficient incident response in SOC environments.